



Physics and Chemistry · 1st year of Bachillerato · The atom and the periodic table · Class 1 of 6 · With April

Mendeleev's gaps



Class video: <https://youtu.be/Ie6CrxaNGao>

The code leads to the video.

Name: _____ Date: _____

While you watch the video

1. What do all the atoms of the same element have in common?
2. With the masses of lithium (6.94) and potassium (39.10), what should sodium have? Compare it with its mass, 22.99.
3. **You, Mendeleev.** Between silicon (28.09) and tin (118.71) an element is missing. How much will it weigh? What will it be like?
4. The first measurement of gallium's density gave 4.7, and the prediction was 5.9. Who do you think was wrong?
5. Why was leaving gaps in the table braver than squeezing in the known elements?

Practice

1. Check the triad of calcium (40.08), strontium (87.62) and barium (137.33).
2. Predict the mass of bromine from those of chlorine (35.45) and iodine (126.90).
3. Cobalt weighs 58.93 and nickel 58.69, yet in today's table cobalt comes first. Why?

Keep thinking

If a theory gets a risky prediction right, do we know it is true? And if it fails just once?